

Token-Level Uncertainty as a Predictor of Hallucination in Large Language Models

A Thematic Literature Review and Research Agenda

Large language models (LLMs) can generate fluent and persuasive text that is nevertheless factually incorrect. This phenomenon, commonly described as hallucination or confabulation, creates a need for reliable signals that can identify unreliable generations before they are trusted or deployed. This review examines token-level uncertainty as one such signal, focusing on token log-probability, predictive entropy, sequence likelihood, perplexity, and related distributional features. The supplied literature-survey spreadsheet contains 32 records, most of which concern LLM fingerprinting and output-distribution analysis rather than hallucination detection directly. Rather than treating these works as direct evidence of hallucination prediction, this review uses them as adjacent evidence that token probability distributions contain stable, informative behavioral signatures. This literature is integrated with research on semantic uncertainty, SelfCheckGPT, relevance-weighted uncertainty, and recent uncertainty-quantification benchmarks. Across these approaches, a consistent pattern emerges: token-level uncertainty can correlate with factual unreliability, but raw uncertainty is not equivalent to factuality. High entropy can reflect legitimate linguistic choice, while a model can assign high probability to a false statement. The strongest direction therefore appears to be calibrated, context-aware and semantically informed token-level prediction, rather than a single global entropy threshold. The review identifies gaps in cross-model calibration, multilingual evaluation, early-token prediction, robustness to decoding settings, domain shift, and causal interpretation of token uncertainty. It proposes future work combining token-level signals with semantic consistency, answer type, retrieval context, and lightweight learned calibration models.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; *Machine learning*; Artificial intelligence.

Additional Key Words and Phrases: Large Language Models, hallucination detection, uncertainty quantification, token-level uncertainty, token entropy, log-probability, predictive uncertainty, semantic uncertainty, calibration, factuality

1 Introduction

Large language models have changed the way natural-language systems are built, but their fluency can conceal an important reliability problem: a model may produce an answer that is grammatically correct, coherent and highly confident in appearance while containing unsupported or false information. Hallucination is therefore not simply a generation-quality problem; it is a problem of estimating when an apparently plausible generation should not be trusted. This distinction is especially important in question answering, education, medicine, law, finance and other settings in which a wrong answer may be more damaging than an incomplete answer.

One promising family of solutions is uncertainty quantification (UQ). In an autoregressive LLM, each generated token is selected from a conditional probability distribution given the prompt and previously generated tokens. If y_t is the token produced at step t , its probability can be written as $p(y_t | x, y_{<t})$. The associated negative log-probability ($-\log p$) measures how surprising the selected token is under the model. At the distribution level, predictive entropy

$$H_t = - \sum_v p(v | x, y_{<t}) \log p(v | x, y_{<t}) \quad (1)$$

Equation 1 measures how dispersed the next-token distribution is. These quantities provide a direct, continuous signal that can be computed during generation when token probabilities are exposed. This token-level view is consistent with the broader uncertainty-quantification literature on autoregressive structured prediction [8].

However, the central research question is not whether uncertainty exists, but whether token-level uncertainty is a useful predictor of hallucination. The distinction matters because language is redundant and many tokens are stylistically interchangeable. A model may be uncertain about whether to say “the study found” or “the study reported” without being uncertain about the underlying fact. Conversely, it may confidently produce a false fact because that fact is linguistically probable or represented in its training distribution. Semantic-entropy research explicitly demonstrates why surface-form uncertainty and meaning-level uncertainty should not be treated as identical [5, 7].

This review is organized around themes, approaches, findings, limitations and research gaps rather than individual paper summaries. The background and conceptual foundation are presented in Section 2. The supplied spreadsheet was treated as the primary literature-survey artifact. It contains 32 records spanning token/output distributions, behavioral signatures, model fingerprinting, phase transitions and related black-box analysis. Only a subset directly concerns uncertainty or hallucination; the remainder is used carefully as adjacent methodological evidence. In particular, the literature on LLM output signatures shows that full next-token distributions can contain substantially richer information than a single response-level score [2], while work on distributional mapping and output-distribution changes further supports the view that token probabilities encode structured information about generation behavior.

2 Background / Conceptual Foundation

2.1 Hallucination as a reliability problem

Hallucination is often used as an umbrella term for generated content that is false, unsupported, or inconsistent with the available evidence. For uncertainty-based detection, the operational definition is more important than the terminology: the detector needs a label indicating whether a generated claim is factually acceptable under a specified reference or evaluation protocol. Hallucination can arise from missing knowledge, ambiguity in the prompt, retrieval failure, reasoning errors, temporal knowledge gaps, or a mismatch between what the model learned statistically and what is true in the world. Consequently, no single uncertainty measure should be expected to identify every failure mode.

2.2 Token-level predictive uncertainty

For a generated sequence $y = (y_1, \dots, y_T)$, the token-level negative log-likelihood is given in Equation 2:

$$\text{NLL}(y \mid x) = - \sum_t \log p(y_t \mid x, y_{<t}). \quad (2)$$

Because longer answers naturally accumulate larger negative log-likelihood, researchers commonly use a length-normalized score, such as mean token log-probability or perplexity. Token entropy instead considers the entire predictive distribution at each step. These measures differ conceptually: log-probability asks how likely the selected token was, whereas entropy asks how uncertain the model was among possible next tokens. A token can have moderate probability even when the rest of the distribution is concentrated, or it can have moderate probability in a very diffuse distribution.

The distinction between token probability and entropy is important for hallucination prediction. A false answer may consist of individually common words and therefore receive high token probabilities. Conversely, a correct answer can contain a rare proper name, number or technical term and therefore produce a low-probability token. Thus, token-level signals should be interpreted as indicators of predictive uncertainty rather than direct truth values.

This workflow is shown in Figure 1.

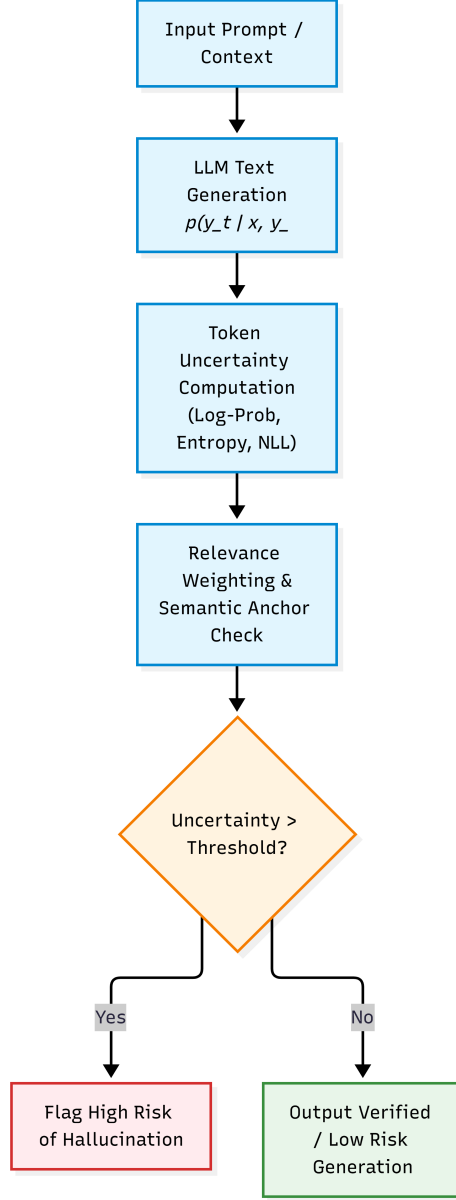


Fig. 1. Token-level uncertainty-based hallucination risk detection workflow.

2.3 From token uncertainty to sequence and semantic uncertainty

A natural extension is to aggregate token-level signals over a sentence or response. Common aggregation functions include mean negative log-probability, maximum token surprise, variance, high-percentile surprise, and weighted averages. Duan et al. [4] argue that tokens contribute unequally to the meaning of a response and propose relevance-weighted uncertainty (SAR/TokenSAR), moving beyond the assumption that every token deserves equal weight.

Semantic uncertainty takes a different route. Instead of asking whether the exact sequence of words is uncertain, it asks whether the model is uncertain about the meaning of its answer. Kuhn et al. [7] introduced semantic entropy by grouping sampled answers according to meaning; Farquhar et al. [5] demonstrated that semantic entropy can be used to detect hallucinations and confabulations. This progression suggests a hierarchy: raw token uncertainty is the most local signal, sequence uncertainty is an aggregation, and semantic uncertainty is a higher-level interpretation.

2.4 Review scope and synthesis strategy

The review uses the supplied literature-survey spreadsheet as its primary survey artifact. The spreadsheet contains 32 records, with a substantial concentration on LLM fingerprinting, provenance, behavioral signatures, and output-distribution analysis. The synthesis therefore distinguishes direct evidence on hallucination or uncertainty from adjacent evidence about the information contained in next-token distributions. Complementary scholarly sources were included when they directly address semantic uncertainty, relevance-aware uncertainty, black-box hallucination detection, or systematic UQ evaluation. The literature is organized thematically rather than chronologically so that methods can be compared by signal, evaluation setting, findings, and limitations. This scope prevents model-fingerprinting studies from being presented as direct proof of hallucination detection while still using their distributional findings to motivate model-conditioned calibration.

3 Thematic Literature Review

3.1 Theme I: Token probabilities as direct confidence signals

The most direct approach uses the probability assigned to the generated token. Mean token log-probability, length-normalized likelihood and perplexity are attractive because they require no additional verifier model and can be computed in a single generation pass. Benchmarking work such as LM-Polygraph places these information-based measures alongside a broader family of white-box UQ methods, making clear that token entropy and token likelihood are foundational rather than isolated metrics [15]. The supplied spreadsheet reinforces this methodological direction: several records study the structure of next-token distributions, and DMAP explicitly treats sequences of next-token probability distributions as a rich representation for downstream analysis [6].

The principal finding across this line of work is that token distributions contain more information than the final decoded text alone. The “LLM Output Signature” work in the spreadsheet is particularly relevant because it proposes learning from the complete sequence of next-token distributions rather than relying only on probabilities of the observed tokens. Its reported gains in hallucination and training-data-contamination detection suggest that the shape of the predictive distribution may carry useful information that a single scalar confidence score discards [2].

Nevertheless, the evidence should not be interpreted as proof that low token probability equals hallucination. The broader UQ literature warns that token-based uncertainty can be miscalibrated and that entropy is not synonymous with factuality. A model can be certain about a false statement because the statement is statistically familiar. This is one of the most important conceptual limitations of the token-level predictor hypothesis.

3.2 Theme II: Entropy and distributional shape

Predictive entropy is attractive because it uses the full next-token distribution rather than only the probability of the selected token. High entropy can indicate that the model sees several plausible continuations, which may occur when it lacks a stable answer. The literature therefore frequently treats entropy as a local uncertainty signal and aggregates it over generated spans.

The supplied spreadsheet contains adjacent evidence from “Phase Transitions in the Output Distribution of Large Language Models,” which demonstrates that distributional changes can be quantified statistically from next-token probabilities [1]. Although this work is not a hallucination detector, it is conceptually important: it supports the idea that generation behavior can undergo measurable changes in distributional structure, suggesting that hallucination prediction may benefit from detecting local shifts rather than relying only on a response-wide mean.

A related contribution is DMAP, which argues that conventional perplexity does not fully capture the context encoded in a conditional distribution [6]. By representing token rank and probability information jointly, DMAP illustrates a broader research trend toward richer distributional descriptors. For hallucination detection, this suggests that features such as entropy, rank, probability curvature, tail mass and changes between consecutive token distributions may complement mean log-probability.

3.3 Theme III: Relevance-weighted token uncertainty

A major limitation of naive token averaging is that natural-language responses contain many tokens that contribute little to factual meaning. Function words, discourse markers and stylistic phrases may be uncertain without indicating factual risk. Duan et al. [4] address this by shifting attention toward relevant tokens and by developing token- and sentence-level uncertainty measures. Their work is important for the present research topic because it changes the question from “How uncertain is the whole response?” to “How uncertain is the model about the tokens that matter for the response’s meaning?”

This approach also suggests a useful interpretation of hallucination. Factual claims frequently depend on a small number of entities, dates, quantities, relations and predicates. If uncertainty is concentrated on those semantic anchors, the signal may be more predictive than an average over all tokens. This hypothesis is plausible but requires systematic evaluation because tokenization can split names and numbers into multiple subword tokens, making the relationship between token uncertainty and claim-level uncertainty nontrivial.

3.4 Theme IV: Semantic uncertainty and self-consistency

Semantic-entropy methods challenge the assumption that uncertainty should be measured at the word or token surface. Kuhn et al. [7] show that two different responses may express the same meaning, so surface-form variation can inflate uncertainty without indicating ignorance. Farquhar et al. [5] extend this idea to hallucination detection, showing that uncertainty over meanings is more relevant to factual reliability than uncertainty over wording.

SelfCheckGPT provides a complementary black-box perspective [9]. Instead of requiring token probabilities, it samples multiple responses and measures whether they agree. The underlying intuition is that known information should produce more consistent samples, while hallucinated claims tend to vary or contradict across samples. The method achieved stronger sentence-level hallucination detection than several baselines in its evaluation, illustrating that output consistency can serve as an alternative uncertainty proxy when token probabilities are unavailable.

Together, semantic entropy and self-consistency reveal a central trade-off. Token-level methods are computationally cheap and temporally local, but can confuse linguistic variability with epistemic uncertainty. Sampling-based methods capture uncertainty at a more semantic level, but require multiple generations and may be expensive. A promising research direction is therefore to use token-level uncertainty as an early warning signal and semantic consistency as a second-stage confirmation mechanism.

3.5 Theme V: Rich output signatures and behavioral distributions

Several works in the supplied spreadsheet investigate LLM fingerprinting and behavioral signatures [3, 10–12, 14]. These papers are not direct hallucination studies, but they provide important evidence for the representation question. LLMmap, natural fingerprinting, behavioral fingerprinting, prefix-tree statistics, and single-token distribution fingerprints all demonstrate that model behavior can be characterized statistically from output distributions. The relevance to hallucination detection is methodological: if model-specific behavior is encoded in fine-grained output probabilities, then hallucination predictors may need to account for the baseline uncertainty profile of the specific model.

This observation is especially important for calibration. Recent fingerprinting work using behavioral and output-distribution signatures further supports the need to account for model-specific baselines [3, 10–12, 14]. A fixed threshold such as “entropy above 2.0 means hallucination” is unlikely to transfer reliably across models with different vocabularies, decoding configurations, sizes, or training procedures. The fingerprinting literature therefore motivates model-conditioned calibration: uncertainty should be interpreted relative to the normal distribution of that model, task and decoding setting.

3.6 Theme VI: Learning from complete token-distribution sequences

The strongest conceptual bridge between the supplied spreadsheet and the assigned topic is the move from handcrafted scores to learned representations of complete output distributions. The LLM Output Signature work uses the sequence of next-token distributions as an input representation and learns a lightweight model for hallucination and contamination analysis [2]. This direction avoids prematurely compressing a response into one number. Temporal patterns—such as sudden drops in confidence, sustained high entropy, repeated uncertainty around entities, or distributional shifts after a reasoning step—can potentially be learned directly.

The limitation is that learned signature models can become dataset-specific. If the classifier learns the lexical or formatting properties of one benchmark, it may appear accurate without learning a general relationship between uncertainty and factual error. Cross-model, cross-domain and cross-task evaluation is therefore essential.

4 Comparison of Existing Approaches

Table 1 compares the major approaches using the dimensions requested for a review synthesis: approach, methodology, dataset or evaluation setting, findings, and limitations. Where a study is primarily methodological or model-characterization work rather than a hallucination benchmark, that distinction is made explicit.

Table 1. Comparative analysis of uncertainty and distribution-based hallucination-related approaches.

Approach	Methodology / signal	Dataset or evaluation setting	Main findings	Key limitations
Token likelihood / perplexity	Mean or normalized log-probability of generated tokens; single-pass white-box UQ.	LM-Polygraph benchmark covering eleven text-generation tasks.[15]	Simple, inexpensive baseline for sequence likelihood.	Can be confidently wrong; affected by length, model and tokenization.
Token entropy	Entropy of the complete next-token distribution at each generation step.	White-box generation settings in which predictive distributions are exposed.	Captures distributional ambiguity that is not visible from the selected token alone.	High entropy can represent harmless linguistic alternatives rather than factual risk.

Approach	Methodology / signal	Dataset or evaluation setting	Main findings	Key limitations
Relevance-weighted uncertainty	SAR/TokenSAR reweights token- and sentence-level uncertainty according to semantic relevance.[4]	Vicuna, WizardLM and LLaMA-2-Chat (up to 33B); reading, science and medical QA.[4]	Relevance-aware UQ improves on uniform token weighting.	Requires relevance estimation and remains sensitive to task context.
Semantic entropy	Samples multiple answers, groups semantically equivalent generations, and computes entropy over meanings.[5, 7]	TriviaQA, SQuAD 1.1, BioASQ, NQ-Open, SVAMP and FactualBio; LLaMA 2, Falcon and Mistral.[5]	Mean AUROC 0.790 versus 0.691 for naive entropy.[5]	Requires multiple generations and semantic clustering/entailment.
SelfCheckGPT	Black-box sampling and agreement/contradiction analysis across multiple generated passages.[9]	GPT-3 + WikiBio; manually annotated factuality.[9]	Higher sentence-level AUC-PR and passage-level factuality correlation than several baselines.[9]	Requires repeated sampling; consistency is not identical to truth.
Output signatures and behavioral distributions	Learns or statistically characterizes sequences of next-token distributions rather than a single scalar score.[2, 3, 10–12, 14]	Gray-box and black-box output-distribution/model-characterization settings; the supplied survey record also reports hallucination and contamination analysis.	Shows that distribution trajectories contain information beyond scalar confidence and that model-specific behavioral structure can be measurable.	Evidence is partly indirect for factuality; learned predictors may become benchmark-, domain-, or model-specific.

Table 1 shows a progression from inexpensive local confidence signals toward semantic and learned distributional representations. Token-level approaches provide the strongest opportunity for online early warning, whereas semantic methods provide a closer connection to meaning but require additional computation. Output-signature and fingerprinting studies add a complementary insight: uncertainty may need to be interpreted relative to a model-specific behavioral baseline rather than against one universal threshold.

5 Research Trends / Developments

The literature shows five major developments. Recent survey work on LLM uncertainty quantification similarly emphasizes the need to organize methods by uncertainty source, evaluation setting and calibration behavior [13]. First, the field is moving from single scalar confidence scores toward richer representations of the token distribution. Second, uncertainty is increasingly treated as multi-scale: token, span, sentence, response and semantic levels can provide complementary signals. Third, relevance weighting is gaining importance because not every token contributes equally to factual meaning. Fourth, researchers increasingly distinguish white-box and black-box settings, recognizing that commercial APIs may not expose log-probabilities. Fifth, calibration and generalization are becoming central concerns as methods are evaluated across models, datasets and application settings.

The spreadsheet’s recent distributional and fingerprinting studies also point toward a sixth development: behavioral statistics can reveal stable structure in LLM outputs even when internal weights are inaccessible. This makes output probabilities valuable not only for uncertainty estimation but also for model characterization. For hallucination research, this creates an opportunity to normalize uncertainty against model-specific behavioral baselines.

6 Challenges and Limitations

6.1 Confidence is not truth

The most fundamental limitation is epistemic. Token probability measures how compatible a token is with the model’s learned distribution, not whether the resulting proposition is true. A false but common statement can be generated with high confidence. Conversely, a correct but rare fact can receive low probability. Therefore, token uncertainty should be described as a predictor of risk, not a direct factuality score.

6.2 Tokenization and linguistic redundancy

Subword tokenization creates measurement artifacts. A single entity may span several tokens, while function words may dominate the token count. This makes raw averaging sensitive to tokenization choices. Semantic entropy partly addresses the underlying issue by moving the uncertainty target from form to meaning.

6.3 Length and aggregation bias

Sequence probability decreases with length, making unnormalized likelihood unsuitable for comparing answers of different lengths. Length normalization helps but introduces its own assumption: every token contributes equally to the expected uncertainty. Max, percentile and weighted statistics may better capture localized risk, but their thresholds require calibration.

6.4 Model and decoding dependence

Different models have different vocabularies, probability sharpness and calibration properties [8, 11]. Temperature, top- p , top- k and other decoding choices also reshape token distributions. A threshold trained on one model may fail on another. The fingerprinting literature makes this issue particularly salient because output distributions can be model-specific even under similar prompts.

6.5 Domain and task dependence

Uncertainty behaves differently in factual question answering, creative writing, summarization and open-ended conversation. A high-entropy creative continuation may be desirable, whereas high uncertainty around a medical dosage or legal citation may be a warning sign. A universal predictor must therefore incorporate task and answer-type context.

6.6 Evaluation leakage and benchmark limitations

Hallucination benchmarks vary in definition, granularity and reference quality. Some labels identify unsupported claims, others identify contradictions, and others use human judgments. If a detector is trained and evaluated on the same style of generated text, it may learn superficial correlations. Robust evaluation requires unseen models, domains, prompt styles and decoding settings.

7 Research Gaps

The literature reveals several gaps directly relevant to the question of whether token-level uncertainty can predict hallucination.

- **Early prediction:** Most evaluations aggregate uncertainty after the full response is available. There is limited systematic evidence on whether the first few high-risk tokens can predict a later hallucinated claim.
- **Claim-level alignment:** Token uncertainty is rarely mapped rigorously to atomic factual claims. A detector may flag a token without identifying which proposition is actually at risk.

- **Cross-model calibration:** Comparable uncertainty thresholds across model families, tokenizers and sizes remain underdeveloped.
- **Decoding robustness:** The effect of temperature, top- p , greedy decoding and constrained decoding on the uncertainty-hallucination relationship needs controlled study.
- **Multilingual validity:** Much evidence is concentrated on English. Tokenization and uncertainty behavior may differ substantially across languages.
- **Domain transfer:** Medical, legal, scientific and everyday factuality have different error distributions and tolerance levels.
- **Causal interpretation:** It remains unclear whether uncertainty is a causal precursor of hallucination or merely a correlated by-product of difficult language generation.
- **Efficient hybrid detection:** There is a need for detectors that combine cheap token-level signals with more expensive semantic verification only when necessary.

8 Future Research Directions

8.1 A calibrated token-risk predictor

A useful next step is a lightweight model that predicts hallucination risk from the token-distribution trajectory. Rather than relying on one threshold, the predictor could use mean and variance of log-probability, entropy, maximum surprise, upper-percentile surprise, entropy change, probability-rank statistics and local distribution divergence. The output should be calibrated so that a predicted 0.8 risk corresponds approximately to an 80% hallucination rate under a clearly defined evaluation setting.

8.2 Semantic-anchor weighting

Token uncertainty should be weighted according to whether a token participates in a factual claim. Entities, numbers, dates, named relations and predicates could receive greater weight than function words. This extends the relevance-aware direction of SAR/TokenSAR while retaining the computational advantage of token-level analysis [4].

8.3 Two-stage early-warning architecture

A practical system could operate in two stages. Stage 1 would monitor token-level uncertainty online and flag suspicious spans. Stage 2 would activate semantic consistency checking, retrieval or an external verifier only for flagged spans. This design could reduce the cost of always-on multi-sample semantic entropy while retaining a stronger second-stage check when uncertainty is concentrated on meaningful content.

8.4 Model-conditioned and task-conditioned calibration

Future studies should report calibration curves separately for each model, domain and decoding configuration. A hierarchical calibration system could first estimate the normal uncertainty profile of a model and then determine whether a new generation is unusually uncertain relative to that baseline.

8.5 Standardized evaluation protocol

A strong benchmark should vary model family, model size, tokenizer, temperature, prompt type, answer length, language and domain. It should report AUROC and AUPRC for detection, expected calibration error for confidence quality, and latency/cost for deployment. Claim-level and token-level localization should be evaluated separately from response-level classification.

8.6 Combining distributional and semantic evidence

The strongest long-term direction is likely not token uncertainty alone. Token distributions, hidden states, self-consistency and external evidence provide different views of reliability [5, 7, 9]. A multimodal uncertainty representation for the text-generation process could learn when each signal should be trusted, while preserving interpretability by showing the high-risk tokens and the evidence behind a final hallucination warning.

9 Conclusion

The literature supports token-level uncertainty as a meaningful but incomplete predictor of hallucination in large language models. Token log-probability, entropy and related statistics provide a direct view of the model’s predictive state and offer an important operational advantage: they can be measured during generation without requiring a separate fact-checking system. The literature survey supplied for this review further indicates that next-token distributions contain rich, structured information about model behavior, strengthening the case for moving beyond simple response-level scores.

At the same time, the evidence does not justify treating uncertainty as a synonym for factual error. Language generation is intrinsically variable, tokenization introduces artifacts, and models can be confidently wrong. Semantic entropy and self-consistency therefore complement token-level approaches by moving the uncertainty target closer to meaning and factual consistency. Relevance-weighted methods provide an intermediate solution by emphasizing tokens that matter more to the response’s semantic content.

The central research opportunity is consequently to transform token uncertainty from a generic confidence measure into a calibrated, claim-aware and model-conditioned hallucination-risk signal. Future systems should detect uncertainty early, localize it to meaningful spans, adapt to model and decoding differences, and selectively invoke semantic or external verification. Such a hybrid approach would preserve the efficiency and interpretability of token-level signals while addressing their most important weakness: a model’s confidence reflects its learned distribution, not necessarily the truth of the world.

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A Literature Survey Synthesis from the Supplied Spreadsheet

The uploaded spreadsheet contains 32 literature records. Most entries concern LLM fingerprinting, provenance and behavioral signatures. For the present review, the records most directly useful for the assigned topic were those addressing output signatures, next-token distributions, distributional changes and behavioral characterization. They were interpreted as adjacent evidence rather than as direct hallucination studies unless the record explicitly stated hallucination detection.

The selected literature-survey records used in the synthesis are presented in Table 2.

Table 2. Selected literature-survey records used in the synthesis

Literature record	Year	Relevance to this review	Use in synthesis
Learning on LLM Output Signatures for gray-box LLM Behavior Analysis	2025	Directly relevant: learns from full next-token distributions for hallucination/contamination analysis.	Evidence for rich token-distribution representations.
One Token Is Enough: Fingerprinting and Verifying Large Language Models from Single-Token Output Distributions	2026	Adjacent: demonstrates model-specific single-token distributions.	Supports model-conditioned calibration.
LLMmap: Fingerprinting For Large Language Models	2025	Adjacent: demonstrates stable behavioral signatures in LLM outputs.	Supports model-specific uncertainty baselines.
Natural Fingerprints of Large Language Models	2025	Adjacent: demonstrates stable behavioral signatures in LLM outputs.	Supports model-specific uncertainty baselines.
Behavioral Fingerprinting of Large Language Models	2025	Adjacent: demonstrates stable behavioral signatures in LLM outputs.	Supports model-specific uncertainty baselines.

Literature record	Year	Relevance to this review	Use in synthesis
Phase Transitions in the Output Distribution of Large Language Models	2024	Adjacent: analyzes statistical changes in next-token distributions.	Motivates temporal/distribution-shift features.
DMAP: A Distribution Map for Text	2026	Highly relevant: represents next-token probability/rank structure.	Supports richer distributional features beyond perplexity.

Note: The spreadsheet was used as the literature-survey input. Because its majority focus is LLM fingerprinting rather than hallucination detection, the review explicitly separates direct evidence from adjacent methodological evidence.